

Autonomy Without Agency: A Formal Distinction Between Autonomous and Agentic AI Systems in Safety-Critical Domains

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Abstract

The AI discourse conflates two fundamentally distinct paradigms, *autonomous* and *agentic* AI, under a shared vocabulary of “AI that acts independently.” This conflation obscures critical architectural differences that carry profound implications for safety, governance, and regulatory compliance, particularly in safety-critical domains such as aviation, industrial control, algorithmic trading, and clinical decision support. This paper provides the first formal, criterion-based distinction between Autonomous AI (Non-Agentic) and Agentic AI, grounded in six architectural dimensions: goal origin, tool selection, reasoning mechanism, self-correction capacity, memory architecture, and operational scope. We formalise the AUTONOMY LOOP as a five-phase closed-loop cycle (Monitor, Decide, Act, Verify, and Loop) and contrast it against the open-ended PPAS (Perceive-Plan-Act-Self-Correct) agent loop from classical agent theory. We systematically map 19 production systems across four sub-types (Continuous Control, Event-Driven, Scheduled/Pipeline, and Autonomous Decision Systems) mapped to an 8-layer technology stack spanning sensor data through governance and oversight. Our empirical survey demonstrates that autonomous systems operating within pre-defined operational envelopes dominate high-assurance domains: algorithmic trading accounts for 60 to 75% of US equity volume, Kubernetes-based auto-scaling underpins approximately 85% of cloud production workloads, and civil aviation autopilots manage 95% of flight time. We quantify seven fundamental limitations of non-agentic autonomy and

map the resulting risk taxonomy against the EU AI Act, GDPR Article 22, MiFID II, FAA/EASA regulations, and FDA Software as a Medical Device (SaMD) frameworks. The formal distinction introduced here provides system designers, regulators, and safety engineers with precise vocabulary to govern the deployment of autonomous systems in high-stakes contexts, and clarifies the boundary conditions under which upgrading a system from autonomous to agentic operation introduces qualitatively new risks. All data and analysis artefacts are publicly available as part of the AI Systems Landscape 2026 repository.

Keywords: Autonomous AI, Agentic AI, Closed-Loop Control, Safety-Critical Systems, Operational Envelope, AIOps, Algorithmic Trading, AI Governance, EU AI Act, GDPR Article 22, Autonomy Loop

1 Introduction

1.1 Background and Motivation

As artificial intelligence increasingly performs consequential actions without direct human authorisation (scaling cloud infrastructure, executing financial trades, moderating social media content, and controlling physical processes), the vocabulary used to describe these systems matters enormously. Terms such as “autonomous,” “agentic,” “self-directed,” and “intelligent” are deployed interchangeably in industry discourse, regulatory filings, and academic literature, creating ambiguity about what level of autonomy a given system actually possesses, what risks it presents, and which regulatory frameworks apply.

This conflation is not merely academic. When regulators ask whether a system is “autonomous,” they are asking whether it requires human-in-the-loop approval for each decision, a question with direct legal consequences under GDPR Article 22 [European Parliament and Council, 2016], the EU AI Act [European Parliament and Council, 2024], MiFID II algorithmic trading provisions [European Parliament and Council, 2014], and FDA Software as a Medical Device guidance [U.S. Food and Drug Administration, 2019]. When safety engineers ask the same question, they are asking about failure mode characteristics, blast radius, and recovery time, all of which differ fundamentally between autonomous and agentic architectures.

1.2 The Core Conflation

The AI field currently uses “autonomous” to describe at least two architecturally distinct paradigms:

Autonomous (Non-Agentic) AI Systems that execute pre-defined policies within constrained operational envelopes, without human approval for individual decisions, but operating within boundaries set entirely by humans at design time. Examples include aircraft autopilots, Kubernetes Horizontal Pod Autoscalers, high-frequency trading engines, and SCADA-controlled industrial processes.

Agentic AI Systems that receive high-level goals, decompose them into sub-tasks, dynamically select and invoke tools, maintain persistent memory across reasoning steps, and self-correct from failures, all within operator-defined guardrails but without pre-scripted decision policies. Examples include LLM-based coding agents [Mahdi, 2026a], research agents, and multi-agent enterprise workflows.

These two paradigms differ not in degree of autonomy but in kind: autonomous systems execute policy; agentic systems reason about which policy to follow. This distinction has profound implications for verification, testing, safety analysis, and regulation.

1.3 Research Objectives

This paper addresses the conflation through five research questions:

- RQ1** What formal criteria distinguish Autonomous (Non-Agentive) AI from Agentive AI along architectural dimensions?
- RQ2** How does the 5-phase Autonomy Loop differ from classical agent loop models, and what are the implications for safety analysis?
- RQ3** What are the dominant sub-types of Autonomous AI in production, and how do they map to a common 8-layer technology stack?
- RQ4** What fundamental limitations does the non-agentive autonomy model impose, and how do these limitations create safety guarantees absent in agentive systems?
- RQ5** How do existing regulatory frameworks map to autonomous versus agentive systems, and where do governance gaps exist?

1.4 Contributions

Our main contributions are:

1. A formal six-dimension criterion set for distinguishing Autonomous AI from Agentive AI, with operational tests for each criterion.
2. Formalisation of the AUTONOMY LOOP as a five-phase closed-loop control cycle and its comparison against PPAS and classical agent models.
3. A comprehensive taxonomy of four Autonomous AI sub-types mapped to an 8-layer technology stack with 30+ platform examples.
4. An empirical characterisation of production autonomous systems across six industries, including quantitative adoption metrics.
5. A seven-point risk taxonomy for autonomous systems with mitigations and a regulatory mapping across five major frameworks.
6. A boundary condition analysis identifying the precise conditions under which autonomous-to-agentive upgrades introduce qualitatively new safety risks.

1.5 Paper Organisation

Section 2 reviews related work and classical control theory. Section 3 formalises the autonomous/autonomous distinction. Section 4 presents the Autonomy Loop. Section 5 maps the 8-layer stack. Section 6 analyses sub-types. Section 7 surveys platforms and tools. Section 8 presents industry use cases. Section 9 covers benchmarks and metrics. Section 10 provides market analysis. Section 11 analyses risks. Section 12 covers regulation. Section 13 identifies open problems. Section 14 concludes.

2 Background & Related Work

2.1 Classical Control Theory and Autonomy

The intellectual foundations of Autonomous AI predate digital computing. Wiener's cybernetics [Wiener, 1948] established the closed-loop feedback paradigm: a system that measures output, compares it to a reference, and adjusts input to minimise error. This principle underlies every autonomous system discussed in this paper, from aircraft autopilots (which maintain altitude by adjusting control surfaces based on barometric sensor feedback) to cloud auto-scalers (which adjust instance counts based on CPU utilisation metrics).

PID (Proportional-Integral-Derivative) controllers [Åström and Hägglund, 2006] operationalised closed-loop control in industrial processes, becoming the dominant technology in the Distributed Control Systems (DCS) that manage chemical plants, power stations, and manufacturing lines. The critical property of PID control, and of all classical autonomous systems, is *predictability*: given the same input state, the system produces the same output action, and the action space is finite and fully specified at design time.

2.2 Autonomy in Computing Systems

IBM's Autonomic Computing manifesto [Kephart and Chess, 2003] introduced four self-* properties: self-configuring, self-healing, self-optimising, and self-protecting, as the target for enterprise IT systems. This vision directly anticipates modern cloud auto-scaling and AIOps platforms, which implement all four properties within pre-defined operational envelopes. The autonomic computing framework explicitly distinguishes between the *policy* that defines what the system should do and the *autonomic elements* that execute that policy, a distinction central to our formal criterion set.

2.3 SAE Levels of Driving Automation

The SAE International standard J3016 [SAE International, 2021] provides the most widely deployed taxonomy of autonomous system capability, defining six levels of driving automation (0 to 5) based on who performs the dynamic driving task, who monitors the driving environment, and who is the fallback performer. SAE levels 0 to 2 involve human performance of the driving task (with varying levels of driver assistance); levels 3 to 5 involve Automated Driving System (ADS) performance. Crucially, even SAE Level 5 (full driving automation in all conditions) does not make a vehicle *agentic*: the system still executes a pre-defined policy (drive safely to the destination) rather than reasoning about what goal to pursue or dynamically selecting which tools to use.

2.4 Agent Theory

Russell and Norvig [Russell and Norvig, 2021] define an agent as “anything that can be viewed as perceiving its environment through sensors and acting upon that environment through actuators.” Under this broad definition, an aircraft autopilot and a GPT-4o-based coding agent are both agents. The distinction lies in the policy type: the autopilot follows a closed-form control law; the LLM agent synthesises plans using general-purpose reasoning. The BDI (Belief-Desire-Intention) model [Rao and Georgeff, 1995] captures the more restricted notion of a *goal-directed* agent that maintains explicit beliefs about the world, desires representing goals, and intentions representing committed plans. Our formal criterion set operationalises this distinction into testable properties.

2.5 Related Surveys

Wang et al. [Wang et al., 2024] survey LLM-based autonomous agents but use “autonomous” to mean both agentic and non-agentic systems. Duan et al. [Duan et al., 2022] survey autonomous systems in physical domains without distinguishing the agentic dimension. To our knowledge, no prior work provides a formal, multi-criterion boundary between autonomous and agentic AI applicable across domains. The companion paper in this series [Mahdi, 2026b] introduces a unified 19-type taxonomy; the present paper deepens the autonomous/agentic boundary within that taxonomy.

3 Formal Distinction: Autonomous vs. Agentic AI

3.1 The Six Architectural Dimensions

We propose six dimensions along which Autonomous (Non-Agentic) AI and Agentic AI differ fundamentally. For each dimension we provide a formal criterion and an operational

test.

Table 1: Six-dimension formal distinction between Autonomous (Non-Agentive) AI and Agentive AI.

Dimension	Autonomous (Non-Agentive)	Agentive
Goal Origin	Pre-defined at design time by humans. The system executes a fixed objective; it cannot modify its own goal.	Self-decomposed from a high-level human objective. The system formulates sub-goals at runtime.
Tool Selection	Fixed. The system uses a specific, pre-configured set of actions (APIs, actuators, control outputs) defined at design time.	Dynamic. The system selects tools from an available set based on reasoning about the current task.
Reasoning Mechanism	Policy execution. Decisions follow a pre-specified mapping (rule, model, controller) from state to action. No open-ended reasoning.	Multi-step planning. The system synthesises novel plans by reasoning about goals, context, and available tools.
Self-Correction	Pre-defined fallback behaviour. Recovery follows scripted contingency logic specified at design time.	Dynamic re-planning. The system diagnoses failure and generates a new plan; recovery is not pre-scripted.
Memory Architecture	Task-scoped state. Persistent state is limited to the variables required for the current operational loop; no episodic memory across tasks.	Persistent episodic memory. The system accumulates experience across tasks and uses it to inform future reasoning.
Operational Scope	Narrow and formally bounded. The operational envelope is explicitly specified; out-of-envelope conditions trigger escalation or safe-fail.	Open-ended within guardrails. The system can pursue novel approaches to achieve its goal within operator-defined constraints.

3.2 Operational Tests

The following operational tests allow practitioners to classify a system along each dimension:

1. **Goal Origin Test:** Can the system modify its own objective without human intervention? *Yes*: potentially agentive; *No*: autonomous.

2. **Tool Selection Test:** At runtime, does the system choose which tools/actions to invoke, or execute a pre-specified action sequence? *Dynamic selection:* agentic; *pre-specified:* autonomous.
3. **Reasoning Test:** Can the system solve a problem type it was not explicitly programmed for? *Yes:* agentic; *No:* autonomous.
4. **Self-Correction Test:** Does the system generate novel recovery strategies at runtime, or follow a pre-scripted fallback? *Novel recovery:* agentic; *scripted:* autonomous.
5. **Memory Test:** Does the system's behaviour change based on experience from previous, distinct tasks? *Yes:* agentic; *No:* autonomous.
6. **Scope Test:** Is the system's operational envelope formally specified and finite? *Yes, finite:* autonomous; *open-ended:* agentic.

Proposition 1 (Formal Criterion). *A system S is classified as Autonomous (Non-Agentic) if and only if it satisfies all six autonomous criteria simultaneously: pre-defined goals, fixed tool selection, policy-based reasoning, scripted self-correction, task-scoped memory, and formally bounded operational scope.*

Note that real-world systems may occupy intermediate positions. A cloud auto-scaler is purely autonomous; an LLM-powered DevOps agent is purely agentic; a reinforcement-learning trading system that learns new strategies from market experience occupies an intermediate position. Our classification is not binary in practice but provides clear poles for anchoring analysis.

3.3 Safety Implications of the Distinction

The autonomous/agentic distinction carries direct safety implications:

- **Formal verifiability.** Autonomous systems with finite, pre-specified action spaces are amenable to formal verification methods (model checking, Lyapunov stability analysis, safety envelopes). Agentic systems with open-ended reasoning are not, in general, formally verifiable.
- **Failure mode predictability.** In autonomous systems, failure modes can be enumerated at design time and tested exhaustively. In agentic systems, failure modes include unbounded plan generation and emergent tool-use patterns that cannot be fully anticipated.

- **Blast radius control.** Autonomous systems operate within a formally bounded authority envelope; blast radius is bounded by design. Agentic systems require guardrails to constrain blast radius, and those guardrails may be circumvented by novel reasoning strategies.
- **Regulatory compliance.** Autonomous systems in regulated domains can demonstrate compliance through conformance testing against specifications. Agentic systems require runtime monitoring and probabilistic compliance assurance, which existing regulatory frameworks do not yet fully accommodate.

4 The Autonomy Loop

4.1 Formal Definition

We formalise the Autonomous AI operational cycle as the AUTONOMY LOOP: a five-phase closed-loop process that enables continuous, unsupervised operation within a pre-defined operational envelope.

Definition 1 (Autonomous System). *An autonomous system $\mathcal{A} = \langle \mathcal{S}, \pi, \mathcal{E}, \mathcal{B}, \mathcal{L} \rangle$ is a tuple where $\mathcal{S}$ is a state space representing the observable environment, $\pi : \mathcal{S} \rightarrow \text{Act}$ is a pre-specified decision policy, $\mathcal{E}$ is an operational envelope (a bounded subset of $\mathcal{S}$), $\mathcal{B}$ is a fallback behaviour triggered when $s \notin \mathcal{E}$, and $\mathcal{L}$ is an immutable audit log.*

Definition 2 (Autonomy Loop). *Given a system $\mathcal{A}$ and an environment state $s_t \in \mathcal{S}$, the Autonomy Loop executes iteratively:*

1. **MONITOR:** $s_t = \text{observe}(\mathcal{S}, t)$: continuously collect telemetry, sensor readings, event streams, and derived metrics.
2. **DECIDE:** $a_t = \pi(s_t)$: apply the decision policy to determine the appropriate action; if $s_t \notin \mathcal{E}$, invoke $\mathcal{B}$ (escalate or safe-fail).
3. **ACT:** execute a_t within the authority boundary (no human approval required).
4. **VERIFY:** $v_t = \text{verify}(s_{t+1}, a_t)$: confirm the action achieved its intended effect; detect anomalous outcomes.
5. **LOOP:** $\log(\mathcal{L}, s_t, a_t, v_t)$; return to Step 1.

The loop repeats indefinitely; humans set boundaries $(\mathcal{E}, \mathcal{B}, \pi)$ but do not participate in individual iterations.

Table 2: Comparison of control and agent loop models.

Model	Phases	Memory	Open Policy	Self-Correction
PID Controller [Åström and Hägglund, 2006]	1	None	No	Via feedback
Autonomic Computing [Kephart and Chess, 2003]	4	System state	No	Via self-healing
OODA Loop [Boyd, 1987]	4	Implicit	Limited	Linear
BDI Agent [Rao and Georgeff, 1995]	3	Explicit	Yes	Via reasoning
Autonomy Loop (Ours)	5	Task-scoped	No	Via $\mathcal{B}$ (fallback)
PPAS (Agentic) [Mahdi, 2026a]	4	Episodic	Yes	Dynamic

4.2 Comparison with Classical Agent Loop Models

4.3 TikZ Diagram

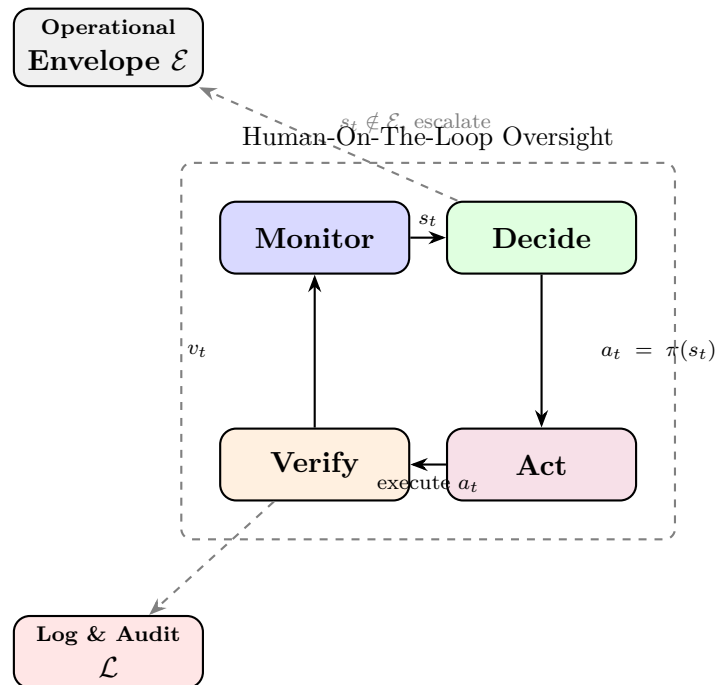


Figure 1: The Autonomy Loop: five-phase closed-loop cycle. Humans define policy π , envelope $\mathcal{E}$, and fallback $\mathcal{B}$ at design time; individual loop iterations require no human participation. Dashed edges show the out-of-envelope escalation path and audit logging.

4.4 Key Autonomy Loop Parameters

5 The 8-Layer Autonomous AI Technology Stack

We decompose the Autonomous AI architecture into eight functional layers, each representing a distinct concern in system design. Unlike the agentic stack (which is organised around reasoning and tool use), the autonomous stack is organised around the closed-loop control cycle and the governance structures that make unsupervised operation safe.

Table 3: Key operational parameters of the Autonomy Loop.

Parameter	Description
Operational Envelope $\mathcal{E}$	Formal bounds within which the system may act autonomously
Decision Policy π	Algorithm (rule, ML model, PID controller) mapping state to action
Authority Level	Maximum magnitude of action the system may take without human approval
Monitoring Frequency	How often state s_t is observed; determines reaction latency
Escalation Thresholds	Conditions that trigger transition to fallback behaviour $\mathcal{B}$
Fallback Behaviour $\mathcal{B}$	Response to out-of-envelope states (safe-fail, hold, escalate to human)
Action Limits	Hard constraints on action magnitude (e.g., max trade size, max scale step)
Audit Trail Depth	Volume of history retained in $\mathcal{L}$ for compliance and forensics

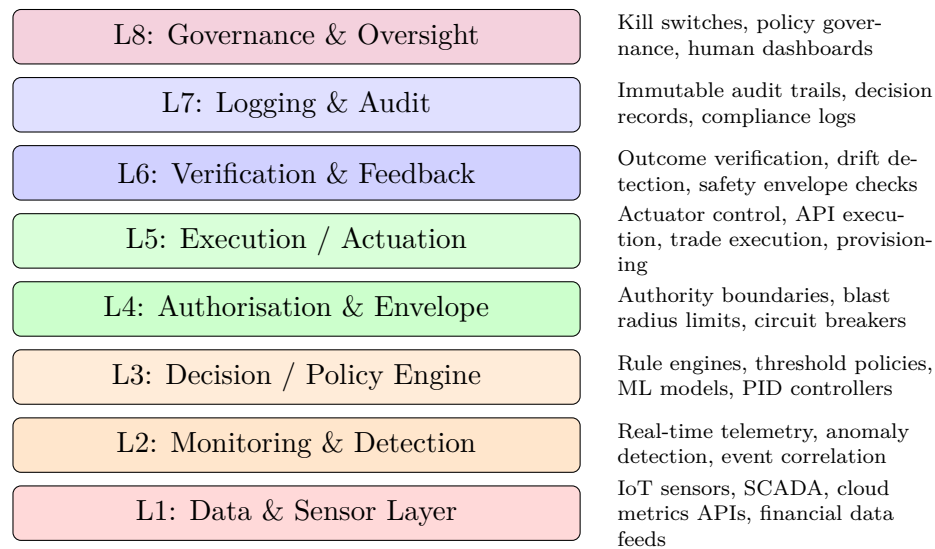


Figure 2: The 8-layer Autonomous AI technology stack. Data flows upward (sensor to decision to actuation); governance flows downward (policy to envelope enforcement to logging).

5.1 Layer-by-Layer Analysis

L1: Data & Sensor Layer. Every autonomous system begins with observation. In industrial settings, this layer comprises physical sensors (temperature, pressure, flow, vibration) connected via SCADA systems or Distributed Control Systems (DCS). In cloud computing, it comprises metrics APIs (CPU, memory, request rate, error rate) surfaced by Prometheus or CloudWatch. In financial trading, it comprises market data feeds (NYSE, NASDAQ, CME), order book data, and alternative data streams. The key engineering constraint at this layer is *latency*: high-frequency trading requires sub-microsecond data delivery; aircraft autopilots require sensor fusion at >100 Hz; cloud auto-scalers operate on 60-second metric windows.

L2: Monitoring & Detection. Raw sensor data is aggregated and analysed to detect conditions that warrant action. This layer implements threshold monitoring, statistical

Table 4: The 8-Layer Autonomous AI Stack with representative technologies.

Layer	Name	Function	Representative Technologies
L1	Data & Sensor Layer	Real-world observation	IoT sensors, SCADA, Kafka, Bloomberg Terminal
L2	Monitoring & Detection	State assessment and alerting	Datadog, Dynatrace, Grafana, PagerDuty
L3	Decision / Policy Engine	State-to-action mapping	Rule engines, PID controllers, Boost, OPA (Open Policy Agent)
L4	Authorisation & Envelope	Authority boundary enforcement	Kubernetes RBAC, rate limiters, circuit breakers (Resilience4j)
L5	Execution / Actuation	Autonomous action execution	Kubernetes HPA/VPA, AWS Auto Scaling, Ansible, Traefik
L6	Verification & Feedback	Outcome checking and loop closure	Health checks, canary deployments, SLO/SLA tracking
L7	Logging & Audit	Immutable decision records	OpenTelemetry, Elasticsearch, immutable S3 logging
L8	Governance & Oversight	Human-on-the-loop control	Review dashboards, kill switches, policy management, SOAR

anomaly detection, and event correlation. Modern AIOps platforms (Datadog, Dynatrace, New Relic) apply ML at this layer to reduce false-positive alert rates, but the action triggered by a detected condition is still governed by the pre-defined policy at L3.

L3: Decision / Policy Engine. This is the cognitive core of the autonomous system, but it is emphatically not open-ended reasoning. The policy π is fully specified before deployment and determines the action for each state. Policy implementations vary by domain: PID controllers in process automation, rule engines (Drools, OPA) in business automation, pre-trained ML classifiers in fraud detection, and discrete-event state machines in CI/CD pipelines. Critically, the policy cannot generate novel strategies; it can only apply its learned or programmed mapping.

L4: Authorisation & Envelope. The authorisation layer enforces the operational envelope $\mathcal{E}$, ensuring that no action exceeds the authority boundary set by humans. This is implemented via rate limiters (maximum API calls per second), position limits (maximum trade size), blast radius controls (maximum number of pods that can be scaled in a single step), and circuit breakers (Resilience4j, Netflix Hystrix) that halt autonomous activity when downstream services exhibit elevated error rates.

L5: Execution / Actuation. Authorised actions are executed at this layer. In cloud computing, Kubernetes HPA adjusts pod replica counts; AWS Auto Scaling terminates and launches EC2 instances. In industrial control, DCS controllers send setpoint com-

mands to actuators (valves, motors, pumps). In algorithmic trading, order management systems (OMS) route orders to exchanges via FIX protocol. The execution layer is designed for determinism and reliability: the same authorised action must produce the same effect every time.

L6: Verification & Feedback. After execution, the system verifies that the action achieved its intended effect. In cloud systems, this involves checking that new instances have passed health checks before routing traffic. In industrial control, it involves confirming that the actuator reached the target setpoint within tolerance. In trading, it involves confirming order fills and updating position records. Verification outputs feed directly back to L2, closing the loop.

L7: Logging & Audit. Every decision $(s_t, \pi(s_t), a_t, v_t)$ is recorded in $\mathcal{L}$ for compliance and forensics. Regulatory requirements (MiFID II, GDPR, FDA audit trail requirements) mandate that this log be tamper-proof and queryable. Modern implementations use append-only object storage (S3 with Object Lock) or distributed ledgers for financial applications.

L8: Governance & Oversight. The governance layer provides the human-on-the-loop interface: dashboards for monitoring system performance, kill switches for emergency halts, policy management tools for adjusting π and $\mathcal{E}$, and periodic review workflows for detecting policy drift or bias accumulation. This layer does not participate in the real-time Autonomy Loop but can modify all lower-layer parameters and halt the system at any time.

6 Sub-Types of Autonomous AI

We identify four operationally distinct sub-types of Autonomous AI, differentiated by triggering mechanism, timing, and decision structure.

6.1 Continuous Control Autonomy

Continuous control represents the oldest form of autonomous AI, implemented in aircraft autopilots since the 1930s and industrialised in DCS and PLC (Programmable Logic Controller) platforms throughout the 1970s to 1990s. Modern implementations maintain strict real-time requirements: aircraft attitude control systems operate at >100 Hz; power grid frequency controllers must respond within 500 ms to maintain 50/60 Hz stability.

The defining safety property of continuous control systems is the *stability certificate*: Lyapunov stability analysis [Lyapunov, 1892] can prove, for many control laws, that

Table 5: Four sub-types of Autonomous AI with defining characteristics and examples.

Sub-Type	Trigger	Timing	Examples
Continuous Control	Continuous feedback signal (sensor reading vs. setpoint)	Real-time / millisecond loop	Aircraft autopilot, cruise control (Automotive), power grid balancing, DCS process control, spacecraft attitude control
Event-Driven	Discrete event or threshold crossing	On-demand / sub-second to minutes	Cloud auto-scaling, automated incident response, circuit breakers, self-healing infrastructure, automated failover
Scheduled / Pipeline	Time trigger or upstream completion event	Scheduled / hours to days	CI/CD pipelines, ETL data pipelines, MLOps retraining pipelines, report generation, backup archival
Autonomous Decision Systems	Application request or market event	Per-decision / microseconds to seconds	Algorithmic trading, dynamic pricing, content moderation, automated loan decisioning, automated underwriting

the system will remain within the operational envelope $\mathcal{E}$ under specified disturbance conditions. This formal verifiability is a key safety advantage over agentic systems.

6.2 Event-Driven Autonomy

Event-driven autonomous systems act when a monitored metric crosses a pre-defined threshold. Kubernetes Horizontal Pod Autoscaler (HPA) is the canonical example: when average CPU utilisation across a deployment exceeds 70%, HPA adds pod replicas; when utilisation falls below 50%, replicas are reduced. Cloud providers report that approximately 85% of production cloud workloads employ auto-scaling [Amazon Web Services, 2024], making event-driven autonomy the most widely deployed form of AI-driven automation.

Self-healing infrastructure, where Kubernetes replaces failed pods and re-routes traffic automatically, has been shown to reduce Mean Time to Recovery (MTTR) by up to 80% compared to manual remediation [Netflix Technology Blog, 2024]. This improvement is

achieved without any reasoning or planning: the system simply detects a pod-health-check failure and triggers the pre-defined replacement policy.

6.3 Scheduled / Pipeline Autonomy

CI/CD pipelines, MLOps retraining workflows, and ETL data pipelines represent a form of autonomy where the trigger is temporal or graph-based (downstream completion) rather than metric-based. The GitLab State of DevOps 2024 report [GitLab Inc., 2024] documents adoption at 78% of software development teams. These systems are autonomous in the sense that they execute multi-step workflows (build, test, deploy, validate) without human approval at each step. However, they remain firmly non-agentic: the workflow graph is static, defined in declarative pipeline configuration (YAML, HCL, Python DAGs), and cannot be modified at runtime.

6.4 Autonomous Decision Systems

Autonomous decision systems represent the highest-stakes category: systems that make consequential decisions (trade, approve, remove, price) in real time without human review. Algorithmic trading constitutes 60 to 75% of US equity market volume [TABB Group, 2024], with high-frequency trading engines operating at nanosecond latencies beyond human reaction time. Content moderation at Meta [Meta Platforms, 2024] autonomously processes approximately 500 million pieces of content per quarter. Automated loan decisioning has compressed approval times from days to seconds at fintech lenders such as SoFi, Lemonade, and Oscar Health.

These systems carry the highest regulatory exposure. Their decision policies (π) must be fully documented, tested for bias, and, in the EU, subject to human review rights under GDPR Article 22 and the EU AI Act.

7 Platforms, Frameworks & Tools

Table 6: Cloud auto-scaling and self-healing platforms.

Platform	Provider	Key Capability
AWS Auto Scaling	Amazon	EC2, ECS, EKS, DynamoDB; predictive scaling
Azure VM Scale Sets	Microsoft	VM + Azure Autoscale for PaaS services
GCP Managed Instance Groups	Google	Compute Engine + GKE Autopilot
Kubernetes HPA / VPA / CA	CNCF	Pod and cluster-level auto-scaling
HashiCorp Nomad	HashiCorp	Workload scheduling; multi-cloud

Table 7: MLOps and autonomous pipeline platforms.

Platform	Provider	Key Capability
Kubeflow	Google (OSS)	End-to-end ML pipelines on Kubernetes
MLflow	Databricks	Experiment tracking, model registry, deployment
Apache Airflow	Apache (OSS)	DAG-based workflow orchestration
Prefect	Prefect	Python-native workflow orchestration
Dagster	Dagster (OSS)	Asset-based data and ML pipelines
Argo Workflows	CNCF (OSS)	Kubernetes-native workflow engine

Table 8: AIOps and IT operations platforms.

Platform	Provider	Key Capability
Datadog	Datadog	Infrastructure monitoring; ML anomaly detection
Dynatrace	Dynatrace	AIOps; root cause analysis; auto-remediation
New Relic	New Relic	Observability; AI-assisted anomaly detection
PagerDuty	PagerDuty	Incident management; automated remediation workflows
Splunk XSOAR	Splunk	SOAR; security automation playbooks

8 Industry Use Cases

Table 10 summarises the principal production deployments across six industry sectors.

9 Benchmarks & Evaluation Metrics

Unlike agentic systems, autonomous systems are evaluated along three orthogonal axes: *autonomy* (how much it does without humans), *reliability* (how often it does it correctly), and *decision quality* (how well its decisions match expert judgement).

9.1 Domain-Specific Performance Data

Representative performance data from production deployments:

- **Cloud auto-scaling:** AWS predictive scaling reduces over-provisioning by 18 to 45%; Kubernetes HPA achieves 99.9%+ uptime across major deployments.
- **Self-healing:** Netflix Chaos Engineering demonstrates MTTR reduction of ~80% vs. manual remediation [Netflix Technology Blog, 2024].
- **CI/CD:** Deployment frequency increases of 200+ times and lead time reductions of 99%+ are documented in the DORA State of DevOps report [DORA Research Program, 2024].

Table 9: Industrial control and SCADA platforms.

Platform	Provider	Domain
Honeywell Experion PKS	Honeywell	Refining, chemicals, oil & gas
ABB Ability Symphony Plus	ABB	Power generation, water utilities
Emerson DeltaV	Emerson	Batch control, safety systems
Siemens SPPA-T3000	Siemens	Power plant control
OSIsoft PI (AVEVA)	AVEVA	Real-time operational data infrastructure

- **Aircraft autopilot:** Manages 95% of flight time with fatal accident rate < 0.07 per million departures (IATA 2024).
- **Algorithmic trading:** HFT market-making reduces bid-ask spreads by 15 to 25% and provides continuous liquidity [TABB Group, 2024].
- **Dynamic pricing:** Amazon reprices millions of products daily; airlines achieve 5 to 15% revenue uplift vs. static pricing.

10 Market & Adoption Data

10.1 Growth Trajectories

The AIOps market (self-managing IT infrastructure) is projected to grow from \$5.2B (2024) to \$18.4B (2030) at a 20% CAGR, driven by the increasing complexity of cloud-native architectures and the operational cost of manual incident management. Autonomous trading infrastructure investment continues to grow as regulatory requirements for electronic trading risk controls create barriers to entry that favour established algorithmic participants. Autonomous vehicle technology, while experiencing slower commercial progression than initial projections, has seen Waymo surpass 20 million fully driverless miles, establishing the operational viability of the model.

Key adoption trends:

- **Growing autonomous scope:** Systems are granted increasing authority as operating track records build, resulting in more decisions, higher-value decisions, and broader operational envelopes.
- **AIOps mainstream adoption:** Self-healing, self-scaling infrastructure is now the default configuration for production cloud workloads.
- **Edge autonomy:** Autonomous decision-making is moving to the network edge, enabling sub-millisecond response times in IoT and industrial applications.

- **Regulatory pressure on autonomous decisions:** EU AI Act and GDPR Article 22 are driving investment in explainability and human review mechanisms for automated decision systems.
- **Autonomous-to-agentic transition:** A growing class of systems is augmenting pre-defined autonomous pipelines with LLM reasoning for exception handling, creating hybrid architectures that partially satisfy our formal criterion set.

11 Risks, Limitations & Failure Modes

11.1 Fundamental Limitations

The architectural properties that make autonomous systems safe, namely bounded scope, pre-defined policy, and formal verifiability, are also their fundamental limitations:

11.2 Safety & Operational Risk Taxonomy

11.3 The Autonomy-to-Agency Upgrade Risk

A particularly important risk scenario arises when an autonomous system is upgraded with agentic capabilities, typically by integrating an LLM for exception handling or natural-language goal specification. Such upgrades may inadvertently satisfy one or more of the formal agentic criteria (dynamic tool selection, open-ended reasoning) without the corresponding safety analysis. We propose the following principle:

Proposition 2 (Upgrade Risk Principle). *Any upgrade of an Autonomous (Non-Agentic) system that causes it to satisfy one or more agentic criteria must be treated as a new system introduction for safety, compliance, and regulatory purposes. The safety certificate of the original autonomous system does not extend to the upgraded system.*

12 Regulation & Governance

12.1 Governance Best Practices

Based on the regulatory landscape and production deployment experience, we identify eight governance best practices for autonomous AI systems:

1. **Human-in-the-Loop Escalation:** Define explicit criteria for when the system must pause and request human authorisation; implement and test the escalation path.

2. **Kill Switch:** Mandatory ability to immediately halt all autonomous activity; must be accessible to on-call operators and testable without production impact.
3. **Authority Boundaries:** Formally document the operational envelope $\mathcal{E}$, authority level, and action limits; review and update as the system’s scope evolves.
4. **Decision Logging:** Maintain an immutable audit trail $\mathcal{L}$ recording input state, policy applied, action taken, and outcome; retain for the regulatory minimum period.
5. **Periodic Review:** Scheduled human review of autonomous decisions to detect policy drift, distributional shift, and bias accumulation.
6. **Blast Radius Limitation:** Enforce hard limits on the maximum impact of any single autonomous decision or action cascade.
7. **Shadow Mode Testing:** Operate new or modified autonomous systems in parallel with existing processes before granting live authority; compare decisions to human baseline.
8. **Rollback Capability:** Where technically feasible, implement the ability to reverse autonomous actions (database transaction rollback, infrastructure state rollback, trade reversal).

13 Open Problems & Future Directions

13.1 Formal Verification at Scale

Current formal verification methods (model checking, Lyapunov analysis, runtime verification) scale to relatively simple state spaces. As autonomous systems integrate ML components with high-dimensional feature spaces, formal safety guarantees become computationally intractable. Research in probabilistic verification [Kwiatkowska, 2020] and abstract interpretation offers partial solutions but has not achieved production deployment for safety-critical ML-driven autonomous systems.

13.2 Distribution Shift and Policy Robustness

Autonomous policies are validated against the distribution of states observed during testing. When deployed systems encounter out-of-distribution states (market microstructure changes, novel attack patterns, unexpected sensor correlations), policy behaviour is undefined. Detecting distribution shift before it causes unsafe autonomous actions remains an open research problem.

13.3 The Hybrid Autonomous-Agentic Architecture

A growing class of production systems integrates autonomous pipelines with LLM-based exception handling. These hybrid systems satisfy the autonomous formal criteria for routine operation but invoke agentic reasoning for out-of-envelope conditions. Developing a formal safety framework for such hybrid architectures, and defining the precise boundary of the agentic component's authority, is a critical open problem.

13.4 Regulatory Harmonisation

The EU AI Act, GDPR, MiFID II, and FDA SaMD represent sector-specific regulatory instruments that were developed before the current generation of autonomous AI. Harmonising these frameworks, and developing sector-agnostic standards for operational envelopes, audit trails, and human oversight mechanisms, would reduce compliance uncertainty and enable broader deployment of safe autonomous AI.

13.5 Autonomous AI in Multi-Domain Operations

Modern autonomous systems increasingly span multiple domains: a cloud-native financial trading platform simultaneously operates autonomous infrastructure scaling (technology domain) and autonomous trading (finance domain). The interaction between domain-specific regulatory frameworks and safety requirements for cross-domain autonomous systems has received little systematic study.

14 Conclusion

This paper has established a formal, criterion-based distinction between Autonomous (Non-Agentic) AI and Agentic AI, two paradigms that are routinely conflated in academic and industry discourse. The six-dimension formal criterion set (goal origin, tool selection, reasoning mechanism, self-correction capacity, memory architecture, and operational scope) provides a practical classification tool with direct implications for safety analysis, regulatory compliance, and system design.

The AUTONOMY LOOP formalises the closed-loop operational cycle of autonomous systems as a five-phase process (Monitor, Decide, Act, Verify, and Loop) in which humans define the policy π , operational envelope $\mathcal{E}$, and fallback behaviour $\mathcal{B}$ at design time but do not participate in individual loop iterations. This architectural property, policy pre-specification within a bounded envelope, is both the primary safety guarantee of autonomous systems and their fundamental limitation.

Our survey of production deployments documents the extraordinary scale of autonomous AI in operation today: 60 to 75% of US equity trading volume, 85% of cloud

workloads, 95% of civil aviation flight time, and 10 billion content moderation decisions per quarter. These systems achieve their operational goals with high reliability precisely because they do not reason openly; they execute pre-verified policies within formally bounded scopes.

The formal distinction introduced here has practical consequences for three communities. For *system designers*, it clarifies when a system design requires the safety analysis procedures of a formal autonomous system versus the guardrail design of an agentic system. For *regulators*, it provides vocabulary to distinguish systems amenable to conformance testing (autonomous) from systems requiring runtime monitoring and probabilistic compliance assurance (agentic). For *safety engineers*, it identifies the precise architectural changes that constitute an “autonomous-to-agentic upgrade” and trigger the need for a new safety case.

As AI systems evolve toward hybrid architectures that combine autonomous pipelines with LLM-based reasoning, the boundary formalised here will become increasingly important and, at the same time, increasingly contested. We offer it as a starting point for the community to develop precise, domain-independent standards for the governance of autonomous AI.

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Table 10: Production Autonomous AI deployments across six industry sectors.

Sector	Use Case	Description	Key Examples
Finance	Algorithmic Trading	Fully autonomous trade execution at microsecond speed	Citadel, Two Jane Street
	Dynamic Pricing	Autonomous price adjustment in real time	Amazon, Uber
	Automated Fraud Prevention	Auto-block fraudulent transactions in real time	Visa, PayPal, Radar
	Automated Underwriting	Instant loan and insurance decisioning	SoFi, Lemonade, car Health
Technology	Auto-Scaling	Autonomous compute resource scaling	AWS, Azure, GCP
	Self-Healing Infrastructure	Automatic failure detection and remediation	Kubernetes, NetFlow
	CI/CD Pipelines	Automated build, test, and deploy on commit	GitHub Actions, Jenkins, Lab CI
	Automated Security Response	Detect and contain threats automatically	Palo Alto XDR, Splunk
Energy	Grid Balancing	Autonomous supply-demand balancing with renewables	National Grid, ERCOT
	Smart Building Management	Autonomous HVAC and energy optimisation	Siemens, Schneider Electric
	Wind Farm Optimisation	Autonomous turbine yaw and pitch control	Vestas, GE, Siemens Gamesa
Aviation	Aircraft Autopilot	Autonomous flight management; manages 95% of flight time	Boeing, Airbus
	Air Traffic Flow	Autonomous conflict detection and flow management	EUROCONTROL, FAA ATFM
	Satellite Station-Keeping	Autonomous collision avoidance and attitude control	Starlink, OneWeb
Manufacturing	Lights-Out Manufacturing	Fully automated production; no operators on floor	FANUC, DMG Mori
	Automated Quality Control	Autonomous inspection at production speed	Cognex, Keyence, Landing AI
	Predictive Maintenance Actuation	Auto-schedule maintenance on degradation signal	Siemens MindSphere
Retail	Automated Replenishment	Autonomous inventory re-ordering at threshold	Amazon, Walmart
	Content Moderation	Autonomous removal of policy-violating content	Meta (500M tions/quarter)
	Automated Advertising	Autonomous bid management and placement	Google Smart bidding, Meta Advantage+

Table 11: Autonomy evaluation metrics.

Metric	Definition
Human Intervention Rate	Fraction of decisions or events requiring human intervention
Autonomous Operating Time	Total time operating without human intervention
Escalation Rate	Fraction of events escalated to human operators
Automation Coverage	Fraction of the total workflow automated end-to-end
Mean Time to Autonomous Resolution (MTAR)	Avg. time from event detection to autonomous resolution

Table 12: Reliability evaluation metrics with indicative targets.

Metric	Definition	Target
Uptime / Availability	% of scheduled time system is operational	$\geq 99.9\%$
MTBF	Mean Time Between Failures	Application-specific
MTTR	Mean Time to Recovery (autonomous)	< 5 min (cloud); < 100 ms (control plane)
False Positive Rate	Fraction of unnecessarily triggered actions	$< 1\%$ (critical systems)
False Negative Rate	Fraction of missed events requiring action	$< 0.1\%$ (safety-critical)

Table 13: Decision quality metrics.

Metric	Definition
Decision Accuracy	Fraction of decisions matching expert judgement
Regret Rate	Fraction of decisions later reversed or corrected
Compliance Rate	Fraction of decisions complying with applicable policy and regulation
Latency	Time from event detection to action execution
Revenue / Cost Impact	Financial impact of autonomous decisions (P&L, costs saved)

Table 14: Autonomous AI adoption metrics (2024).

Market Segment	Metric	Source
Algorithmic Trading	60 to 75% of US equity volume	TABB Group
Global AIOps Market	~\$5.2B; CAGR ~20%	IDC / Gartner
Cloud Auto-Scaling Adoption	~85% of cloud workloads	AWS / Azure
CI/CD Adoption	~78% of software teams	GitLab Survey
Automated Content Moderation	~10B actions/quarter (Meta)	Meta Transparency
Dynamic Pricing Adoption	~45% of large retailers	McKinsey

Table 15: Fundamental limitations of Autonomous (Non-Agentive) AI.

Limitation	Description
Fixed Scope	Cannot handle situations outside the defined operational envelope $\mathcal{E}$; out-of-envelope events trigger escalation or safe-fail
No Novel Reasoning	Cannot reason about new problem types; the policy π covers only the situation space anticipated at design time
Policy Rigidity	Follows pre-defined π ; may not respond optimally to unprecedented edge cases or distribution shifts
Cascading Failures	Autonomous actions can cascade across interconnected systems, amplifying errors at machine speed
Lack of Judgement	Cannot exercise contextual judgement in ethically complex or multi-stakeholder situations
Opacity of ML Decisions	When π is an ML model, individual decisions may not be explainable or auditable at the feature level
Over-Reliance Risk	Operators may lose the skills required to perform manual operations when the system fails

Table 16: Safety and operational risk taxonomy for autonomous systems.

Risk	Description	Mitigation
Flash Crash	Cascading autonomous trading amplifies market moves at extreme speed	Circuit breakers, kill switches, position limits
Automation Bias	Operators trust the system excessively and fail to detect or override errors	Regular human review; clear authority boundaries
Cascading Automation	One autonomous action triggers others, creating unexpected chains	System decoupling; rate limiting; blast radius controls
Silent Failure	System operates within $\mathcal{E}$ but makes subtly incorrect decisions	Outcome monitoring; statistical drift detection
Cybersecurity Compromise	Compromise enables autonomous execution of malicious actions	Defence-in-depth; anomaly detection; secure-by-design
Bias Amplification	Autonomous decision systems encode and amplify historical biases at scale	Fairness constraints; bias monitoring; regulatory compliance

Table 17: Regulatory framework mapping for Autonomous AI systems.

Framework	Relevance to Autonomous AI
EU AI Act (2024) [European Parliament and Council, 2024]	Autonomous decision systems in high-risk domains (finance, employment, credit, education, law enforcement) require conformity assessment, technical documentation, and post-market monitoring. Human oversight mechanisms are mandatory for high-risk AI.
GDPR Article 22 [European Parliament and Council, 2016]	Right not to be subject to purely automated decision-making with legal or similarly significant effects. Requires human review mechanism for automated credit, employment, and insurance decisions.
MiFID II / Regulation AT [European Parliament and Council, 2014]	Algorithmic trading systems must implement pre-trade risk controls, kill switches, and circuit breakers. Firms must maintain records of all algorithmic orders for five years.
SEC Rule 15c3-5 [U.S. Securities and Exchange Commission, 2010]	US broker-dealers using algorithmic trading must implement market access risk controls, including order limit checks and erroneous order prevention.
ECOA / Fair Lending Laws [Consumer Financial Protection Bureau, 2023]	Automated credit decisioning must comply with anti-discrimination requirements; adverse action notices must explain the basis for automated rejections.
FDA SaMD Guidance [U.S. Food and Drug Administration, 2019]	Autonomous medical diagnostic and treatment decision support software must be classified as a medical device; higher-risk autonomous clinical decisions require 510(k) clearance or PMA approval.
EASA / FAA Regulations	Autopilot systems require certification under aviation safety standards (DO-178C for software, DO-254 for hardware); operational approval tied to safety case documentation.